

AUTOMATIC WATER PUMP CONTROLLER

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Save water and electricity with this fully automatic water pump controller. The circuit is very simple consisting of mostly switches and relays, so anyone with little knowledge of electronics can

house. It can be extended to more floors by adding a similar module for each floor. In case you want the motor to be controlled only from the main unit, remove the floor module and run the motor directly from the main unit to the tank module by ground-floor module. You can easily remove the



system. The system has been divided into three modules for simplicity and better understanding: Ground-floor module, first-floor module and tank module. Fig. 2 shows the circuit of the water pump controller.

Ground-floor module. The ground-floor module consists of regulated power supply, level indicator and control unit.

The 230V AC mains is stepped down by transformer X1 to deliver the secondary output of 15V, 500 mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitors C1 and C2, then regulated by IC 7812 (IC1). Capacitor C3 bypasses the ripples, if any, present in the regulated 12V power

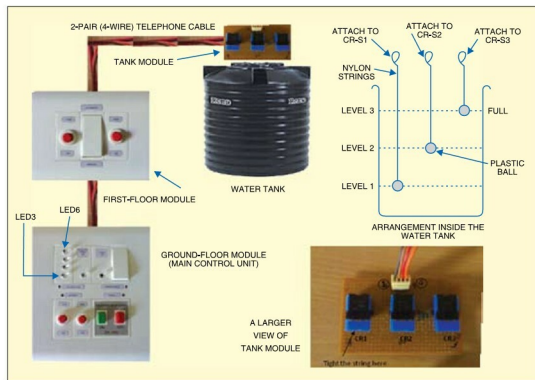


Fig. 1: Block diagram of automatic water pump controller

assemble it easily. If the circuit fails at any time, you can still run the motor by using the manual switch provided on the main control unit. You can also turn on/off the motor irrespective of the water level in the tank by using push-to-on/push-to-off switch.

Another unique feature of this circuit is a first-floor module to control the motor from the first floor of your

floor module at any time as explained in the construction section.

Using some very tricky logic and simple circuitry, the controller can also show you the current level of water in the tank through glowing of the four LEDs.

Circuit description

Fig. 1 shows the block diagram of the automatic water pump controller

supply. LED1 acts as the mains-on indicator and resistor R1 limits the current through LED1.

The level indicator section consists of four LEDs. LED2 indicates that the motor is switched on and LED3 through LED6 show the level of water in the tank.

Three BC547 transistors are used as inverters. When water level goes below level 1, base of transistor T1